A PROJECT TITLE

ON

CHRONIC KIDNEY DISEASE PREDICTION USING XGBOOST AND LIGHTGBM GRADIENT-BOOSTED TREES

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Name of Title: Chronic Kidney Disease Prediction Using XGBoost and LightGBM Gradient-Boosted Trees

1. ABSTRACT:

Chronic Kidney Disease (CKD) is a progressive condition affecting a significant portion of the global population, often remaining undiagnosed until advanced stages due to its largely asymptomatic early progression. Early and accurate prediction of CKD is therefore critical for timely clinical intervention, effective disease management, and improved patient outcomes. This project proposes a machine learning framework for CKD prediction using two state of the art gradient boosted tree algorithms, XGBoost and LightGBM, applied to clinical and laboratory data comprising features such as blood pressure, serum creatinine, haemoglobin, blood urea, albumin, blood glucose, and other renal function indicators. The dataset was pre-processed through handling of missing values, encoding of categorical variables, and feature scaling, followed by exploratory data analysis to identify key predictors of renal dysfunction. Both models were trained and evaluated using standard performance metrics, including accuracy, precision, recall, F1 score, and area under the ROC curve, with hyperparameter tuning performed through grid search and k fold cross validation to optimize predictive performance and reduce overfitting. Comparative analysis reveals that both XGBoost and LightGBM achieve high classification accuracy on the given dataset, with LightGBM demonstrating faster training times owing to its histogram based, leaf wise growth strategy, while XGBoost offers marginally improved robustness on smaller and imbalanced datasets through its level wise tree construction and built in regularization mechanisms. Feature importance analysis further identifies serum creatinine, haemoglobin, and albumin levels as the most influential predictors of CKD onset. The results suggest that ensemble gradient boosting methods provide a reliable, interpretable, and computationally efficient approach for CKD risk stratification, with strong potential for integration into clinical decision support systems for early diagnosis, screening, and preventive care in healthcare settings.

2. INTRODUCTION:

Chronic Kidney Disease (CKD) is a major public health concern, marked by a gradual and often irreversible decline in kidney function. Since early stages typically show no visible symptoms, many cases go undetected until the disease reaches a critical stage requiring dialysis or transplantation, increasing the burden on healthcare systems. With growing availability of clinical data, machine learning offers a promising way to flag at-risk patients earlier. Gradient boosted decision tree algorithms such as XGBoost and LightGBM are popular for their strong predictive performance and efficient handling of large, high-dimensional data. This project applies both algorithms to a CKD dataset, comparing their accuracy, efficiency, and interpretability to support early and reliable diagnosis of kidney disease.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10 / 11 or Linux (Ubuntu 20.04+)
- Programming Language: Python 3.9 or above

- IDE / Notebook: Jupyter Notebook / Google Colab / VS Code
- Libraries: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn
- Dataset: UCI Chronic Kidney Disease Dataset (CSV format)
- Version Control: Git / GitHub

Signature of the Guide **Course Instructor**